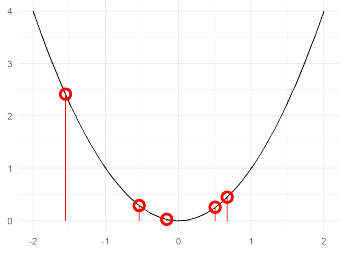
Introduction to Machine Learning

Advanced Risk Minimization
Regression Losses: L2 and L1 loss



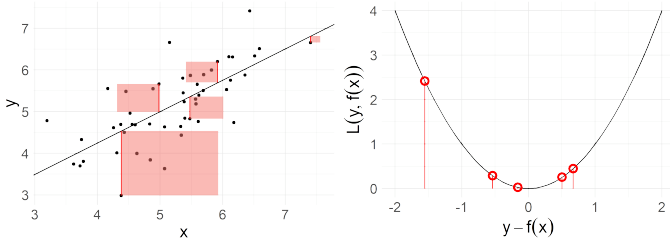
Learning goals

- Derive the risk minimizer of the L2-loss
- Derive the optimal constant model for the L2-loss
- Know risk minimizer and optimal constant model for L1-loss

L2-LOSS

$$L\left(y, f(\mathbf{x})\right) = \left(y - f(\mathbf{x})\right)^2 \quad \text{or} \quad L\left(y, f(\mathbf{x})\right) = 0.5\left(y - f(\mathbf{x})\right)^2$$

- Tries to reduce large residuals (if residual is twice as large, loss is 4 times as large), hence outliers in y can become problematic
- Analytic properties: convex, differentiable $\Rightarrow$ gradient no problem in loss minimization
  (**Warning**: $\mathcal{R}_{\text{emp}}(f)$ can still be non-smooth/non-convex due to $f(\mathbf{x})$)

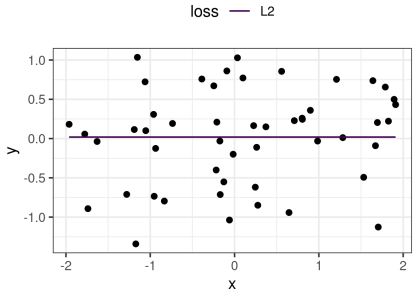


L2-LOSS: OPTIMAL CONSTANT MODEL

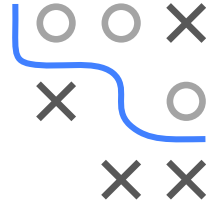
Let us consider the (true) risk for $\mathcal{Y} = \mathbb{R}$ and $L2$-Loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$ with $\mathcal{H}$ restricted to constants. As derived previously, for the squared loss, the optimal constant model is given by

$$f_c^* = \arg\min_{c \in \mathbb{R}} \mathbb{E}_{xy}\left[(y - c)^2\right] = \mathbb{E}_y[y]$$

We also derived: the empirical constant minimizer is $\hat{f}_c = \bar{y}$.



L2-LOSS: RISK MINIMIZER



Let us consider the (true) risk for $\mathcal{Y} = \mathbb{R}$ and the $L2$-Loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$ with unrestricted $\mathcal{H} = \{f(\mathbf{x}) : \mathcal{X} \to \mathbb{R}^g\}$.

Since $\mathcal{H}$ is unrestricted, at any point $\mathbf{x} = \tilde{\mathbf{x}}$, we can predict any value c we want. To find

$$f^*(\tilde{\mathbf{x}}) = \text{argmin}_c \mathbb{E}_{y|x} \left[L(y, c) \mid \mathbf{x} = \tilde{\mathbf{x}} \right] ,$$

we can apply the previous derivation of f_c^* under $L2$ loss using the conditional label distribution. The best point-wise prediction is then given by the cond. mean

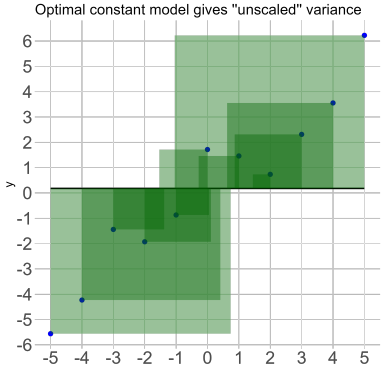
$$f^*(\tilde{\mathbf{x}}) = \mathbb{E}_{y|x} \left[y \mid \mathbf{x} = \tilde{\mathbf{x}} \right] .$$

L2 LOSS MEANS MINIMIZING VARIANCE

Rethinking what we did in the opt. constant model: We optimized for a constant whose squared distance to all data points is minimal (in sum, or on average). This turned out to be the mean.

What if we calculcate the loss of $\hat{\theta} = \bar{y}$? That's $\mathcal{R}_{\text{emp}} = \sum_{i=1}^{n} (y^{(i)} - \bar{y})^2$.

Average this by $\frac{1}{n}$ or $\frac{1}{n-1}$ to obtain variance.



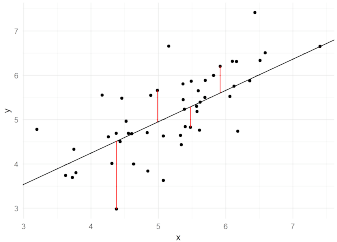
- Generally, if model yields unbiased predictions, $\mathbb{E}_{y \mid \mathbf{x}}[y - f(\mathbf{x}) \mid \mathbf{x}] = 0$, using $L2$-loss means minimizing variance of model residuals
- Same holds for the pointwise construction / conditional distribution considered before

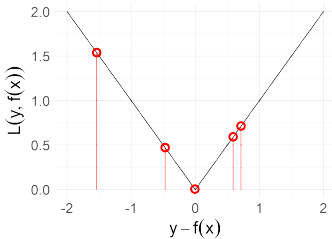
L1-LOSS

The L1 loss is defined as

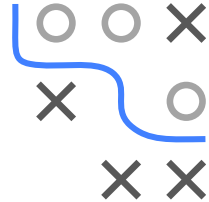
$$L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$$

- More robust than $L2$, outliers in y are less problematic.
- Analytical properties: convex, not differentiable for $y = f(\boldsymbol{x})$ (optimization becomes harder).





L1-LOSS: RISK MINIMIZER



We calculate the (true) risk for the $L1$-Loss $L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$ with unrestricted $\mathcal{H} = \{f : \mathcal{X} \to \mathcal{Y}\}$.

- We use the law of total expectation

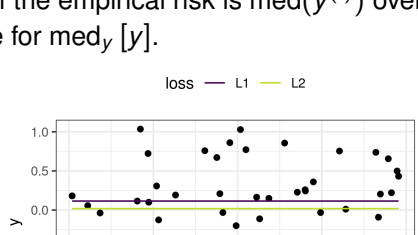
$$\mathcal{R}(f) = \mathbb{E}_x \left[\mathbb{E}_{y|x} \left[|y - f(\mathbf{x})| \; | \mathbf{x} = \tilde{\mathbf{x}} \right] \right].$$

- As the functional form of f is not restricted, we can just optimize point-wise at any point $\mathbf{x} = \tilde{\mathbf{x}}$. The best prediction at $\mathbf{x} = \tilde{\mathbf{x}}$ is then

$$f^*(\mathbf{x}) = \arg\min_c \mathbb{E}_{y|x} \left[|y - c| \right] = \text{med}_{y|x} \left[y \mid \mathbf{x} \right].$$

L1-LOSS: OPTIMAL CONSTANT MODEL

The optimal constant model in terms of the theoretical risk for the L1 loss is the median over y:

$$f_c^* \quad = \quad \text{med}_{y|x}\left[y \mid \mathbf{x}\right] \overset{\text{drop } \mathbf{x}}{=} \text{med}_y\left[y\right]$$

The optimizer $\hat{f}_c$ of the empirical risk is $\text{med}(y^{(i)})$ over $y^{(i)}$, which is the empirical estimate for $\text{med}_y\left[y\right]$.